



MACHINE LEARNING

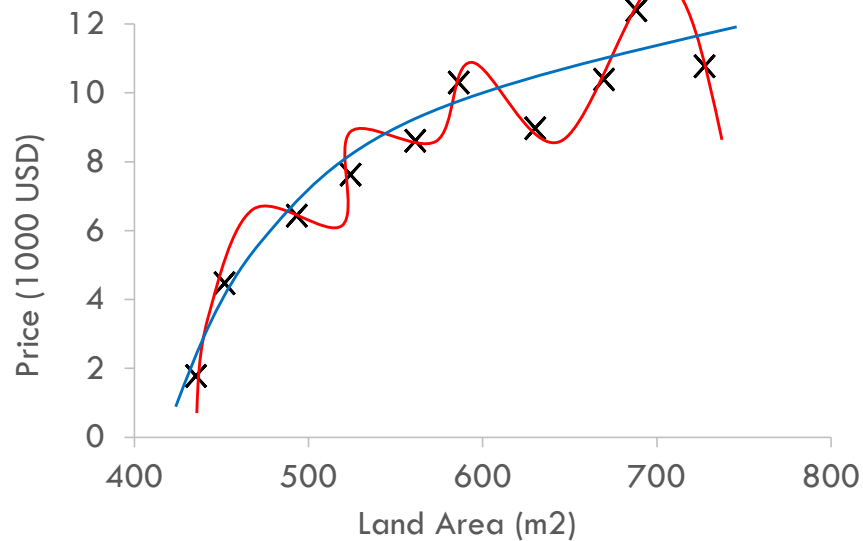
LESSON 05: Model Selection and Evaluation

Prepared by Dr. Dona Valy

Inspired by and adapted from the Coursera Machine Learning course by Andrew Ng
(coursera.org/learn/machine-learning)

HYPOTHESIS EVALUATION

- Overfitting Problem



It fails to generalize to new examples not in the training set

Dataset with many input features?

HYPOTHESIS EVALUATION

Dataset

Size in m ² (x)	Price in 1000 USD (y)
90	7.1
130	10.9
210	19.2
300	28
350	32.8
420	39.9
480	46.1
530	51
640	62.2
710	68.9

Training Set

$$\begin{aligned} &(x^{(1)}, y^{(1)}) \\ &(x^{(2)}, y^{(2)}) \\ &\dots \\ &(x^{(m)}, y^{(m)}) \end{aligned}$$

Test Set

$$\begin{aligned} &(x_{test}^{(1)}, y_{test}^{(1)}) \\ &(x_{test}^{(2)}, y_{test}^{(2)}) \\ &\dots \\ &(x_{test}^{(m_{test})}, y_{test}^{(m_{test})}) \end{aligned}$$

HYPOTHESIS EVALUATION

- Training/testing procedure

- Learn parameter Θ from training data (by minimizing error/cost $J(\Theta)$)

- $J(\Theta) = \frac{1}{m} \sum_{i=1}^m C(h(x^{(i)}), y^{(i)})$

- Compute test set error:

- $J_{test}(\Theta) = \frac{1}{m_{test}} \sum_{i=1}^{m_{test}} C(h(x_{test}^{(i)}), y_{test}^{(i)})$

- In the case of logistic regression/classification problem, a misclassification error can be computed:

- $err(h(x), y) = 1$ if it x is classified incorrectly or 0 otherwise

$$\Rightarrow \text{test error} = \frac{1}{m_{test}} \sum_{i=1}^{m_{test}} err(h(x_{test}^{(i)}), y_{test}^{(i)})$$

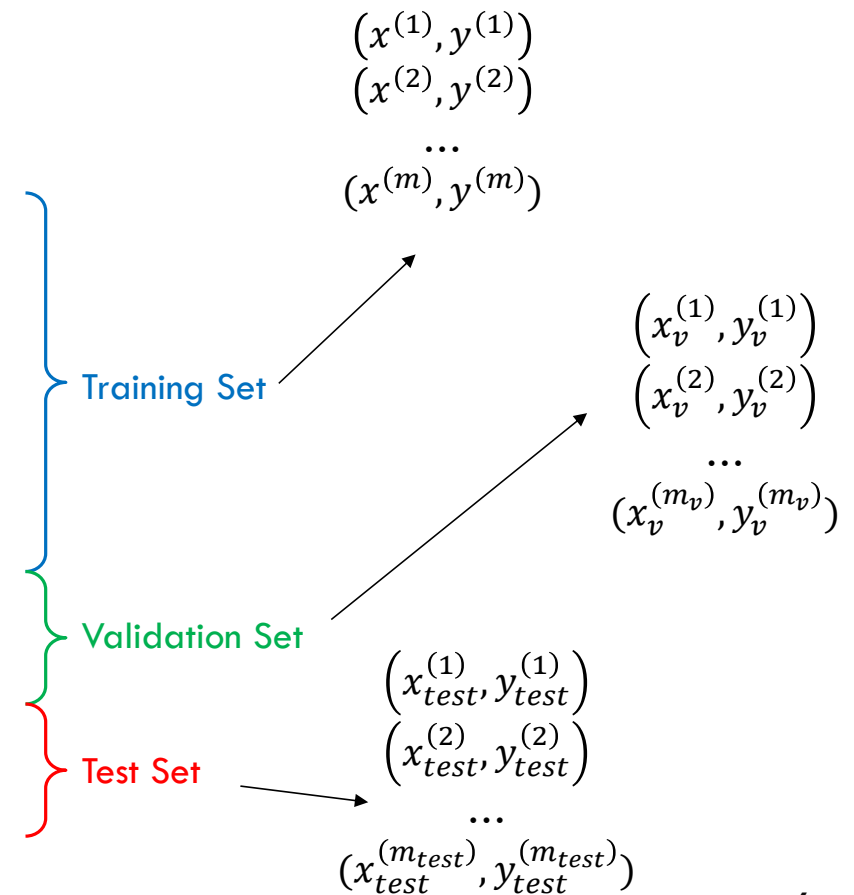
MODEL SELECTION

- Example: we build a neural network with l hidden layers each layer with k units
 - Model 1: $h_1(x), l = 1 \Rightarrow \Theta_1 \Rightarrow J_{test}(\Theta_1)$
 - Model 2: $h_2(x), l = 2 \Rightarrow \Theta_2 \Rightarrow J_{test}(\Theta_2)$
 - Model 10: $h_{10}(x), l = 10 \Rightarrow \Theta_{10} \Rightarrow J_{test}(\Theta_{10})$
- How well does each model generalize?
 - Compute test set error $J_{test}(\Theta_l)$
 - **Problem:** the extra parameter (hyperparameter) l might fit to the test set

MODEL SELECTION

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MODEL SELECTION

- Train/validation/test error
 - Training error: $J(\Theta) = \frac{1}{m} \sum_{i=1}^m C(h(x^{(i)}), y^{(i)})$
 - Validation error: $J_v(\Theta) = \frac{1}{m_v} \sum_{i=1}^{m_v} C(h(x_v^{(i)}), y_v^{(i)})$
 - Test error: $J_{test}(\Theta) = \frac{1}{m_{test}} \sum_{i=1}^{m_{test}} C(h(x_{test}^{(i)}), y_{test}^{(i)})$
- For each model, we minimize $J(\Theta) \Rightarrow J_v(\Theta)$
 - Choose the best model using the validation set
 - Estimate generalization error on test set $J_{test}(\Theta)$

ERROR METRICS FOR SKEWED CLASSES

- Example: Cancer classification
 - Train a binary classification model $h(x)$ ($y = 1$ if the patient has cancer or 0 otherwise)
 - Find out that you got 1% error on test set; however, only 0.5% of the patients have cancer in the test set
 - What if we compare our model with another model which always predict that the patient does not have cancer for any input x ?

PRECISION AND RECALL

- Precision: of all patients where we predict $y = 1$, what fraction actually has cancer?

$$\frac{TP}{TP + FP}$$

- Recall: of all patients that actually have cancer, what fraction is correctly detected as having cancer?

$$\frac{TP}{TP + FN}$$

		Actual Output	
		1	0
Predicted Class	1	True Positive (TP)	False Positive (FP)
	0	False Negative (FN)	True Negative (TN)

F SCORE

- How to compare precision/recall numbers?

	Precision (P)	Recall (R)	Average	F Score
Model 1	0.9	0.2	0.55	0.33
Model 2	0.4	0.7	0.55	0.51
Model 3	0.1	1	0.55	0.18

- F Score: $2 \frac{PR}{P+R}$

EXERCISE 5.1

Look back at the logistic regression model corresponding to $h_2(x)$ in Exercise 3.1.2. Divide the dataset into 60% and 40% for training and test set respectively. Retrain the model on the training set and evaluate it on the test set using precision, recall, and F score metrics.

EXERCISE 5.2

Given a dataset about chemical measurements of wine*, build a neural network as a classifier to predict the quality (Medium, Good, or Excellent) of wine. Divide the dataset into 3 subsets: training (60%), validation (10%), and test (30%). Use the validation set to choose the number of hidden layers (between 2 and 4 hidden layers) and the number of units in each layers (either 25, 50, or 100) for your neural network. Finally compute the error rate of the model that you select on the test set.

* Dataset from <https://machinelearningmastery.com/standard-machine-learning-datasets>